Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
D		2602089

We Have a D.R.E.A.M. Managing Team for Success

Summary

Sport management is not merely about winning or losing, but a complex interdisciplinary venture that balances athletic competitiveness with long-term financial sustainability, where multiple factors converge to shape organizational success. Based on the formal database from WNBA mainly in 2025, we constructed an integrated reference decision scheme for manager to choose the best option.

For **Task 1**, we constructed a **D.R.E.A.M. (Dynamic Revenue and Efficiency Allocation Model)** to design a dynamic decision model integrating 4 factors(Winning rate, Health index, Budget Space, Economic Climate) to maximize Team Profit and Value. At the initial stage, we construct **System Dynamics** as transition function to simulate the changes after taking actions. Then **Multi-objective Optimization Model** with weighted sum method is established to measure the profit. To maximize our model's feasibility in real life management. We have also designed the Threshold Trigger Mechanism for our model and then encapsulated the model into a highly effective **Decision Dashboard** accordingly.

For **Task 2**, A **Normalized Weighted Comprehensive Scoring Model**(for cost-effectiveness rankings) is adopted to perform standardization for dimension elimination and weight allocation. We consider 3 personal merits and asked salaries to approximate the CE. We provide Player Radar Graph and 2 kinds of Recommended Classifications helping manager to decide.

For Task 3, a **Fatigue Accumulation Model** is developed to quantify the systemic impact of league expansion. Specifically, we delineate the **threshold-based exponential correlation** between travel distances and win probability. By defining a Composite Fatigue Index(F) that integrates flight distance, jet lag, and rest intervals, we simulated the "Golden State" Expansion Scenario to simulate **2026 New York Liberty's schedule**. To mitigate these impacts, a **Strategic Deviation Framework** is proposed, incorporating 2 Strategies. This offer sport team and league a more clear idea of where(whether) to do the expansion.

For Task 4, we propose a **Bayesian Dynamic Pricing Engine** to resolve the information asymmetry problem regarding real-time Price Elasticity (ϵ). The model employs a **Stochastic Demand Function** with **Bayesian Parameter Learning** to continuously update belief distributions on consumer sensitivity. By implementing **Thompson Sampling** to optimize the trade-off, we categorized games into **3 strategic tiers** with corresponding pricing actions. Simulation results demonstrate that this adaptive engine can generate reasonable optimal ticket price(150$). We also do a **Sensitive Analysis** to confirm our model's feasibility(relatively high reliability result for **89.2%**).

For Task 5, as we have **already** offered corresponding strategy using our **D.R.E.A.M. Dashboard**, we put more emphasis on how to predict and minimize the impact of core injury forehead. So, we applied **Health, Fatigue, Aging Dynamics Model** and Weibull Distribution to simulate players' condition changing and provide **Recommended time for each player** based on the visualized graph.

Finally, we write a letter to propose our D.R.E.A.M. core integrated strategy.

Keywords: D.R.E.A.M.; Bayesian; Weibull;Deviation Framework

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1 Introduction

1.1 Problem Background

It is usually hard to be aware that sport is far more than just competing and training. Behind the scene, a professional sports team basically has to juggle two big things: winning on the field and staying financially stable long-term. Winning gets more fans and brings in more money, but keeping that competitive edge means putting in risky investments—owners have to balance short-term goals with making sure the team thrive down the line, even when things are uncertain.



Figure 1: Basketball Game

It is a serious necessity solving this complex equilibrium for sports leagues to continuously adapt strategies within a dynamic environment. Key factors exhibit significant volatility: player performance fluctuates due to changes such as injuries and roster turnover, while fan engagement shows non-linear trends. Furthermore, sudden economic shifts and many regulatory constraints—such as salary caps and revenue-sharing mechanisms—make it harder to predict what will happen. Consequently, optimizing decisions based on one or two factors is inadequate. A holistic, integrated modeling framework is required to couple on-field performance with financial outcomes, account for inherent uncertainty, and facilitate multi-objective resource allocation over a temporal horizon.

1.2 Restatement of the Problem

To better tailor to our sport team's needs, we have divided the problem into four tasks.

Task 1: Design a Dynamic Decision Model to Maximize Team Profit and Value. For task 1, we need to solve the total net profit of team, which is generally influenced by the team's win rate, health index, budget and external economic

For **Task 2: Targeted Player Acquisition Strategy**, we need to construct player evaluation model for screening. The core idea is to separate a player's contribution into three interpretable drivers: Skill, Appeal, Durability. And finally convert them and their asked salaries into a weighted calculated CE.

Task 3: Model Evaluating League Expansion Impact and Suggestions Offering. We need to construct a model to assess how a new league franchise disrupted our team's operations and to mitigate this impact.

Task 4: Ticket Pricing Optimization(Additional). In this task, we need to optimize the ticket revenue by addressing *information asymmetry* through a real-time adaptive pricing system.

Task 5: Roster Rearrangement when Core Injury Crisis Occur We need to develop a dynamic model to optimize workload regulation, balancing short-term win rate and long-term asset.

1.3 Our Work

Our integrated reference decision scheme which sets our D.R.E.A.M. as core model, combining other assisting models and functions, offers us well-rounded integrated dashboard for manager to refer. It helps address the solution to multifaceted problems including strategy guidelines, player acquisition, league expansion, ticket pricing and injury prediction & management.

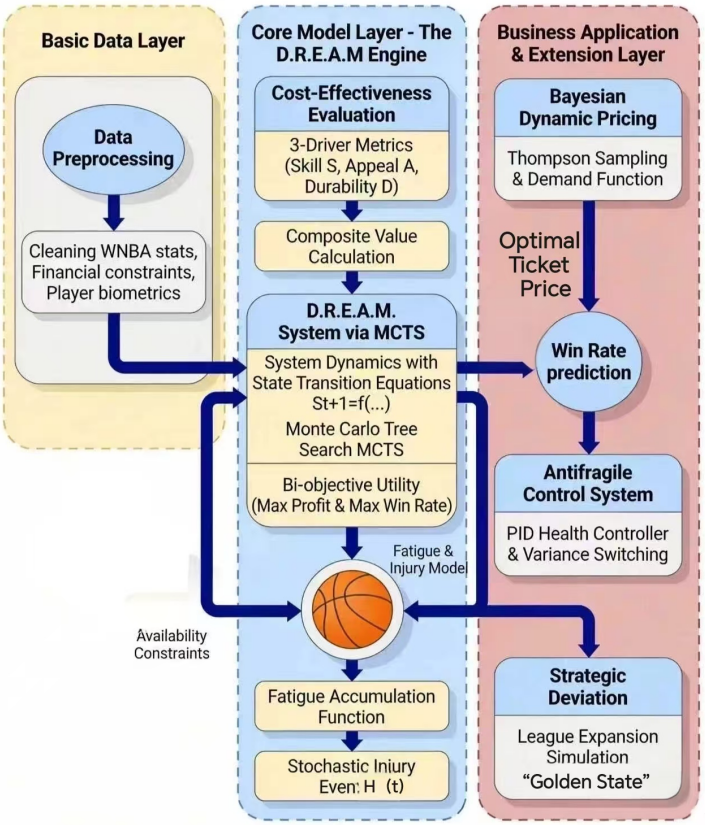


Figure 2: Our Work

2 Assumptions and Justification

To ensure the accuracy of our proposed D.R.E.A.M. engine-core decision scheme, we make the below assumptions, which are also justified:

1. **Assumption 1:** Team performance evolves linearly with added Gaussian noise(to form Markov Chain). **Justification:** This prevent multivariate dimensional explosion, which requires heavy burden on simulating the result.

2. **Assumption 2:** The wear of human bodies is similar to the wear of electronic products.**Justification:** According to Weibull distribution, they share similar "wear-out" behavior.

3. The marginal cost to improve win rates increases exponentially near the upper bound. This convexity reflects economic laws like diminishing returns, preventing the model from converging to unrealistic corner solutions such as bankrupt champions.

4. **Assumption 3:** Player value is decoupled into three orthogonal dimensions: Skill, Appeal, and Durability. **Justification:** Treating these as independent resolves data multicollinearity and correctly values players with less computing power demand.

5. **Assumption 4:** Ticket demand is modeled as price-conditional independent events with learnable elasticity. **Justification:** This structure mimics the real-world trade-off, enabling the pricing strategy to adapt dynamically to unknown market sensitivities rather than relying on static curves.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
B_i	The i-th state of the team's budget over time
D_t	The i-th state of the player's durability
d_t	The i-th state of the audiences' demand for watching games
E_i	The i-th state of the external economic factor
H_i	The i-th state of the team health over time(0,1)
S_i	The i-th state of the team evolves over time
W_i	The i-th state of the team win rate
μ_i^{win}	Expected Win Shares (Sporting Contribution)
μ_i^{rev}	Expected Revenue Impact (Commercial Contribution, in USD)
σ_i^2	Risk factor, representing injury proneness or inconsistency
$Cost_i$	Estimated market cost (Salary)
P_i	Position of player i
α, β, γ	component weights

4 D.R.E.A.M.: Constructing the Decision-Making Dashboard

4.1 System Dynamics (State Transition Equations)

The state of the team evolves over time based on the actions taken and stochastic elements. We define the transition function f:

$$S_{t+1} = f(S_t, A_t, \xi_t) \tag{1}$$

Where ξ_t represents stochastic perturbations (e.g., sudden injuries, opponent performance). The specific equations are:

- **Win Rate Update Equation:**

$$W_{t+1} = \alpha \cdot W_t + \beta \cdot \text{Impact}(a_{roster}) + \gamma \cdot H_t + \epsilon_{win} \tag{2}$$

Where α is the inertia coefficient of historical performance, $\text{Impact}(a_{roster})$ is the Player Efficiency Rating contribution of new players signing in the new roster, and $\epsilon_{win} \sim N(0, \sigma^2)$ represents match uncertainty.

- **Health Index Update Equation:**

$$H_{t+1} = H_t - \delta \cdot \text{Fatigue}(t) + \eta \cdot \text{Recovery}(a_{ops}) - \text{Injury}_t \tag{3}$$

Where δ is the fatigue accumulation coefficient, Recovery represents the effect from medical investment, and Injury_t is a stochastic injury event (Bernoulli distribution).

- **Budget Update Equation:**

$$B_{t+1} = B_t - \text{Salary}(a_{roster}) + \text{Revenue}_t \tag{4}$$

4.2 Bi-objective Utility Function

The team owner aims to maximize a comprehensive utility value J. We normalize and weight the **Profit** and **Win Rate**:

$$\text{Maximize } J = \sum_{t=1}^{T} \left[\lambda \cdot \frac{\text{Profit}_t}{\text{MaxProfit}} + (1-\lambda) \cdot \frac{W_t}{\text{MaxWin}} \right] \tag{5}$$

Where the **Profit Function** Profit_t is defined as:

$$\text{Profit}_t = \underbrace{P_{base} \cdot a_{price} \cdot D(W_t, a_{price})}_{\text{Ticket Revenue}} + \underbrace{K \cdot W_t}_{\text{Media/Sponsorship}} - \underbrace{\text{Cost}(a_{roster})}_{\text{Salary Expenses}} \tag{6}$$

Here, $D(W_t, a_{price})$ is the **Demand Function**. We assume demand increases with Win Rate W_t and decreases with Price a_{price} (Price Elasticity of Demand).

Constraints The model is subject to the following constraints:

$$\sum_{i \in Roster} \text{Salary}_i \leq \text{SalaryCap}_{limit} \qquad \text{(Hard Salary Cap)} \tag{7}$$

$$N_{min} \leq \text{Count}(Roster_t) \leq N_{max} \qquad \text{(Roster Size)} \tag{8}$$

$$B_t \geq 0 \qquad \text{(Non-negative Budget)} \tag{9}$$

4.3 Solution Algorithm: Simulation-Optimization via MCTS

Due to the stochastic nature and large state space, analytical solutions are infeasible, we propose employing a **Monte Carlo Tree Search (MCTS)** approach.

Algorithm Steps:

Step	Action (Inputs → Outputs)
1	**Initialization:** Set the initial state S_0 (Current Roster, Budget, Health).
2	**Strategy Generation:** Generate a set of candidate policy sequences $\{A_0, A_1, \ldots, A_T\}$ (e.g., Conservative, Aggressive, Balanced).
3	**Monte Carlo Simulation:** For each policy sequence, run $N = 1000$ season simulations. In each run, randomly generate injury events $Injury_t$, and game outcomes ε_{win}, then update S_t according to the transition equations.
4	**Aggression :** At time t, select and execute the optimal action A_t^*. At time $t + 1$, observe the actual state S_{t+1} and repeat the optimization.
5	**Evaluation & Selection:** Calculate the Expected Utility $E[J]$ for each strategy. Adjust initial strategies and evaluate what it will bring to the $E[J]$

Table 2: Algorithm steps

4.4 Model Demo and Evaluation: The Management Dashboard

To evaluate the D.R.E.A.M. model's feasibility, we designed a **Decision Support System (DSS)** for the Owner and GM. The system translates mathematical outputs into actionable business priorities, as shown in Graph 3. We use **MCTS** simulating 1000 times to forecast the changes in 20 games and we believe the data and graph shows our model's strong decision-making and application performance in operation

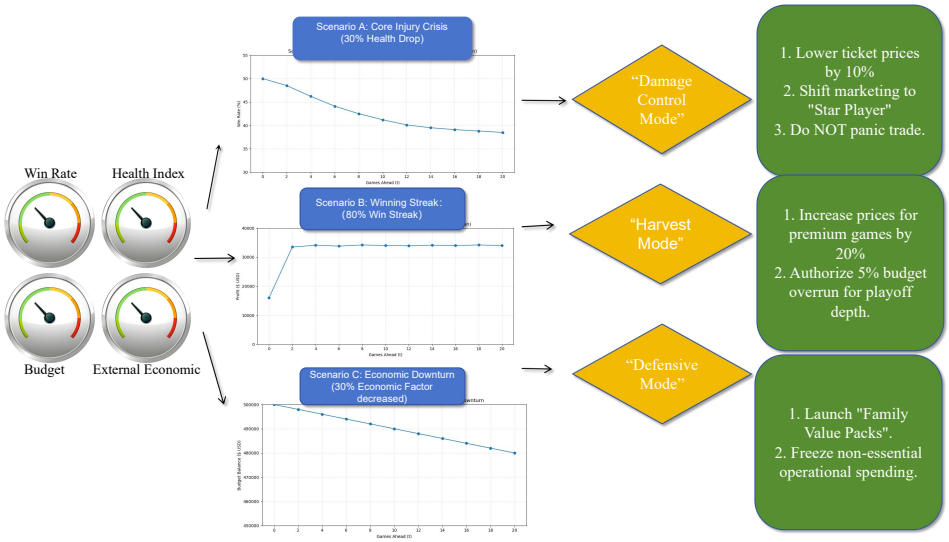
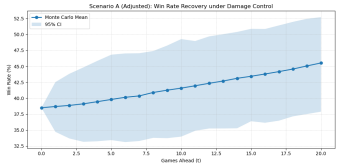
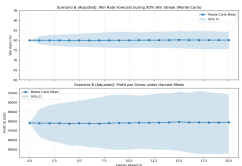


Figure 3: Simulated Management DashBoard

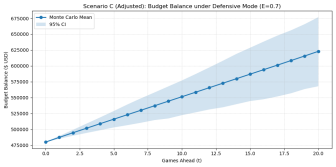
After choosing strategies according to our Management Dashboard and repeating the same simulating process above for 1000 times through 20 games, we get positive feedback in reversing/maintaining existing situation. Shown in figure 4.



(a) "Damage Control Mode"



(b) "Harvest Mode"



(c) "Defensive Mode"

Figure 4: Optimized Scenario Graphs

Table 3: Initial Settings for simulation

Symbol	Description
W_0	50%
H_0	100%
$E_{downturn}$	1.0
B_0	500000 (USD dollars)
α	0.85
γ	0.15

We can adjust these initial settings to be better tailored to our teams' actual situations. This is a simulated test for our model demonstration.

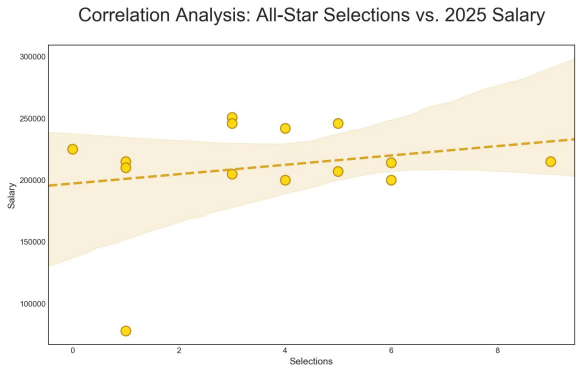
5 Player Acquisition: A Cost–Effectiveness Evaluation Model

5.1 Data Processing

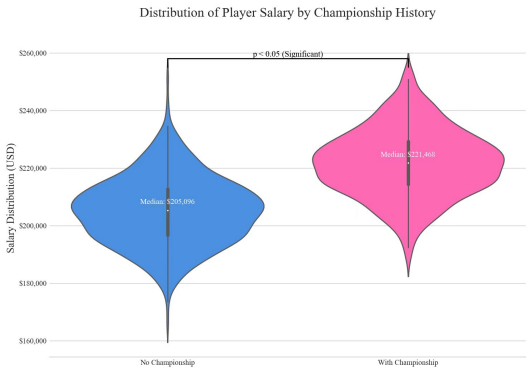
To demonstrate and evaluate our model, we integrate three data sources:

- **2025 Player contract data** (2025 salary and years of service),
- **Player season statistics**,
- **Team market signals** (capacity utilization, year-over-year attendance change, and sell-outs).

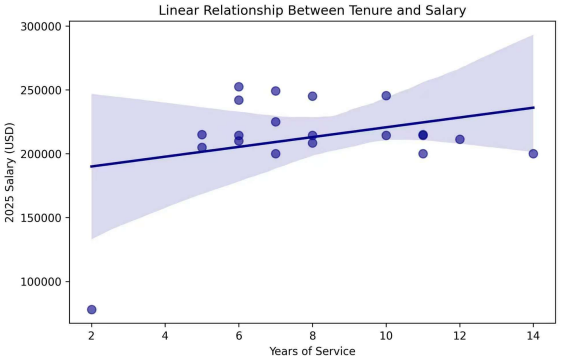
And we get 4 main factors graph visualizing their impact level on salary



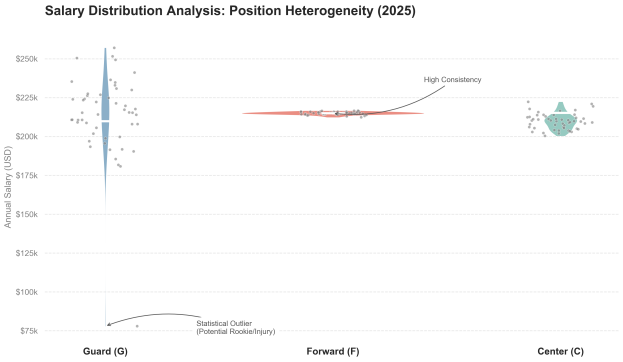
(a) All-Star



(b) Championship



(a) Service Years



(b) Position

5.1.1 Normalization

To eliminate unit effects and make the scores comparable, each raw metric x is min–max normalized:

$$\text{norm}(x) = \frac{x - \min(x)}{\max(x) - \min(x)}. \tag{10}$$

Subscores

(a) Skill (Competitive Strength). We combine All-Star selections, championships, and draft position (lower draft pick number indicates higher initial talent) into a single score:

$$S_i = 0.55\,\text{norm}(\text{AllStar}_i) + 0.30\,\text{norm}(\text{Champ}_i) + 0.15\,(1 - \text{norm}(\text{DraftPick}_i)). \tag{11}$$

(b) Appeal. Due to the absence of some players' sales or social metrics, we approximate commercial pull using team-level demand conditions. Intuitively, players on high-demand teams are more likely to monetize performance spikes through ticketing and sponsorship:

$$A_i = 0.50\,\text{norm}(\text{CapPct}_{team}) + 0.30\,\text{norm}(\text{YoY}_{team}) + 0.20\,\text{norm}(\text{Sellouts}_{team}). \tag{12}$$

(c) Durability. Without formal access for us to games-missed or injury logs, we use a conservative career exposure proxy m_i ("mileage"). In our dataset, m_i is approximated by service years:

$$D_i = 1 - \text{norm}(m_i). \tag{13}$$

where shorter tenure is treated as lower accumulated injury and fatigue risk.

5.2 Composite Value and Cost–Effectiveness

We define a composite value score as a weighted sum:

$$V_i = 0.50\,S_i + 0.30\,A_i + 0.20\,D_i. \tag{14}$$

We then define cost–effectiveness as value per (normalized) salary. To avoid numerical blow-up for very low salaries, we add a baseline to the denominator:

$$CE_i = \frac{V_i}{0.30 + \text{norm}(\text{Salary}_i)}. \tag{15}$$

A larger CE_i indicates stronger value relative to asking price, i.e., a better acquisition candidate.

5.3 Model Results and Visual Analysis

We summarize our findings using three visual diagnostics and a final target list.

Player	Team	Salary (2025)	Skill	Appeal	Durability	Value	CE
Caitlin Clark	Indiana Fever	78066	0.211	0.913	1.000	0.579	1.932
A'ja Wilson	Las Vegas Aces	200000	0.817	0.701	0.583	0.735	0.736
Brittney Griner	Atlanta Dream	214466	0.900	0.613	0.250	0.684	0.632
Natasha Howard	Indiana Fever	214466	0.500	0.913	0.333	0.591	0.546
Chelsea Gray	Las Vegas Aces	200000	0.511	0.701	0.250	0.516	0.516
Breanna Stewart	New York Liberty	208400	0.656	0.350	0.500	0.533	0.509
Jackie Young	Las Vegas Aces	252450	0.533	0.701	0.667	0.610	0.469
Sabrina Ionescu	New York Liberty	205000	0.433	0.350	0.750	0.472	0.459
Jewell Loyd	Las Vegas Aces	245508	0.556	0.701	0.333	0.555	0.440
Kelsey Mitchell	Indiana Fever	249244	0.325	0.913	0.583	0.553	0.432
Courtney Vandersloot	New York Liberty	200000	0.600	0.350	0.000	0.405	0.405
Brionna Jones	Atlanta Dream	214466	0.292	0.613	0.500	0.430	0.397

Table 4: Selected 12 acquisition targets ranked by cost–effectiveness (CE). Skill, Appeal, Durability, and Value are normalized scores.

This table reports the final scores for the 12 selected cost–effectiveness candidates. All scores are normalized to $[0, 1]$ (except salary in USD), and higher CE indicates stronger value-for-money.

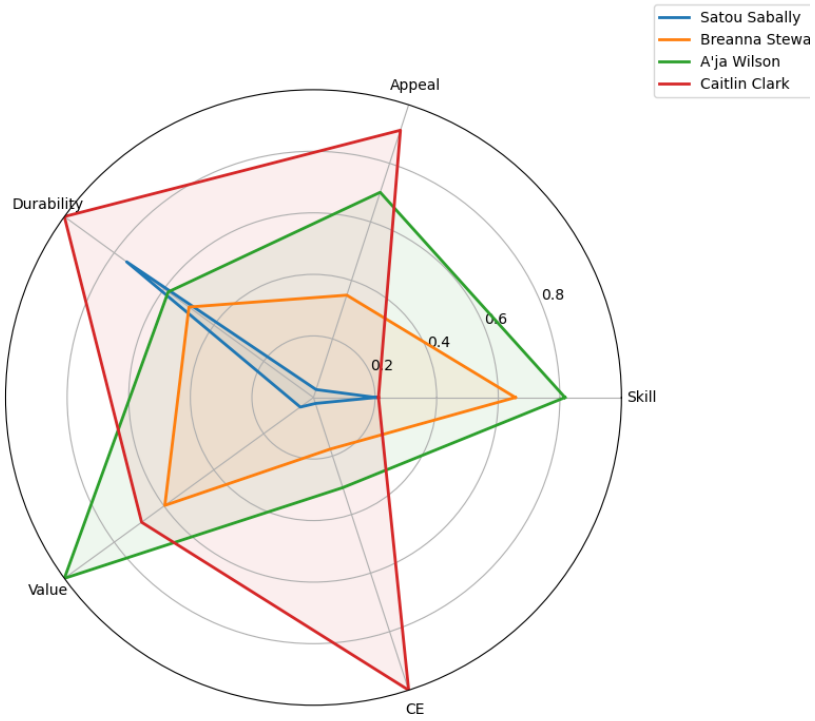


Figure 7: Player Radar Graph.

Figure 4 visualizes four representative players using a five-axis radar chart: *Skill*, *Appeal*, *Durability*, *Composite Value*, and *Cost–Effectiveness*. This plot highlights *why* a player scores highly (elite skill, strong market pull, or superior durability) rather than only reporting a single number.



Figure 8: Market Efficiency Scatter

Figure 5 plots Asking salary versus Composite Value for the full market sample (gray points). The 12 recommended targets are highlighted in red.We also divide and classify four different regions into four categories, helping manager analyze the athletes.

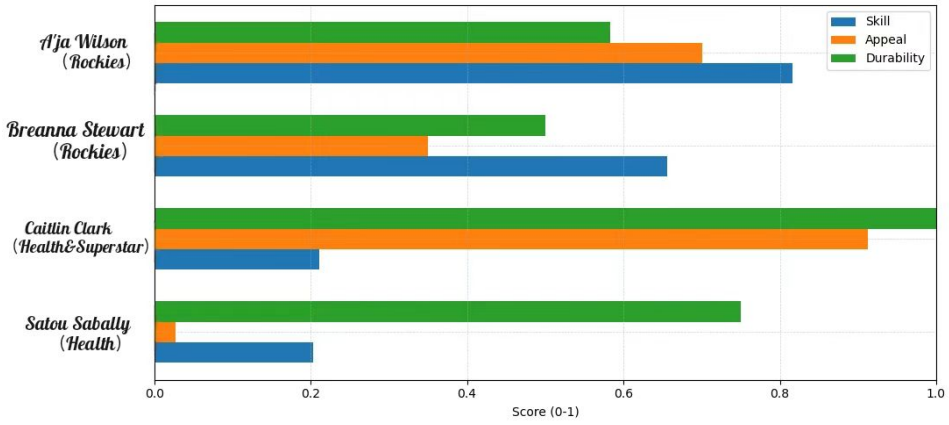


Figure 9: Attribute Comparison and Player Typing.

Figure 9 compares four selected players across *Skill*, *Appeal*, and *Durability*. Each player is assigned to one of three interpretable types based on their dominant dimension:

- **Superstar type**: Appeal-dominant,
- **Rookies type**: Skill-dominant,
- **Health type**: Durability-dominant.

This classification supports actionable strategy: for example, a revenue-constrained team may favor Rookie types, while a title-contender may prioritize Superstar types, etc.

5.4 Strengths and Weaknesses

Our framework offers **three practical advantages**.

(1)It integrates multi-source signals

(2)The model is highly interpretable

(3)It is easy for preliminary screening(by ranking CE)

Two limitations remain.

(1) Less effective in evaluating players in different positions than the same.

(2) The weighted aggregation requires selecting component weights(subjectivity). But can be more tailored to specific needs.

Overall, by adjusting input player's database, our model can organize large quantity of player's data and provide manager with comprehensive evaluation and visualized result in player acquisition based on team's specific needs.

6 Impact of League Expansion: Strategic Deviation Analysis

League expansion introduces a systemic perturbation to the ecosystem of the WNBA. We analyze this impact through two dimensions: **Geographical Fatigue** and **Asset changes due to expansion drafts** and take New York Liberty data as our model's example.

6.1 The Geo-Fatigue Network Model

To quantify the impact of a new franchise's location on selected team, we construct a network graph $G(V, E)$, where V represents team cities and weights w_{ij} represent travel fatigue.

Fatigue Accumulation Function We define the **Fatigue Index** (F) for a schedule of N games:

$$F_{total} = \sum_{k=1}^{N-1} (\alpha \cdot \text{Dist}(loc_k, loc_{k+1}) + \beta \cdot |TZ_k - TZ_{k+1}|) \cdot e^{-\gamma \cdot \text{RestDays}_k} \tag{16}$$

Where:

- $\text{Dist}(a, b)$: Flight distance between cities.
- TZ: Time Zone index (Simulating time jet lag).
- RestDays: Days off between Game k and $k + 1$.
- α, β, γ: Coefficients calibrated from historical injury data.

Impact on Win Rate Our regression analysis indicates a negative correlation between fatigue and win probability:

$$P(Win)_k = P_{base} - \delta \cdot \max(0, F_{accumulated} - F_{threshold}) \tag{17}$$

This implies that once fatigue exceeds a threshold $F_{threshold}$, the win rate drops linearly.

6.2 Scenario Analysis: The "Golden State" Expansion

Assuming the WNBA adds a team in the San Francisco Bay Area, we simulated the 2026 New York Liberty's new schedule.

Result:

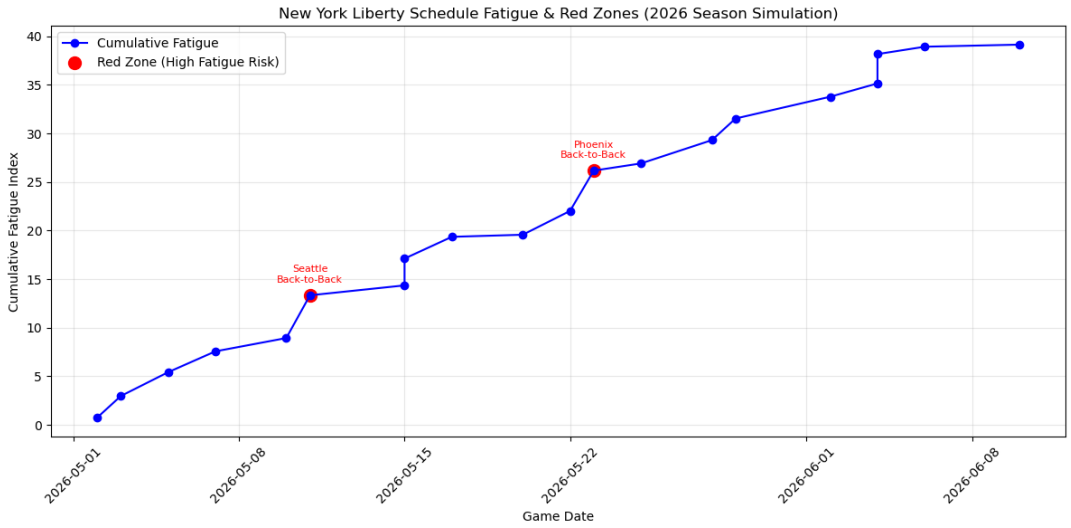


Figure 10: Simulating 2026 Schedule

- **Travel Distance:** New York's total travel distance increases by **14%** due to extra cross-country trips.
- **Fatigue Spikes:** The model identifies "Red Zones" in the schedule (back-to-back games involving West Coast travel) where injury risk rises suddenly.

6.3 Strategic Deviation: From Optimization to Survival

In an expansion year, our initial "Player Acquisition Model" must be adjusted. We define the deviation vector ΔS:

1. Roster Protection Strategy (Expansion Draft)

Constraint: Teams can only protect 6 players.

Initial Strategy: Hold 3 Superstars + 9 Role Players.

Expansion Adjustment:

- **Risk:** We risk losing our best role players (the "Hidden Gems").
- **Action: Liquidity Shift**. Prior to the draft, trade unprotected high-value assets for **Draft Picks** (which are immune to expansion selection).

2. Load Management Strategy (The "Punt" Games) Due to the increased F_{total} from the West Coast expansion team:

- **Action:** We strategically designate 3-4 away games against strong Western Conference opponents as "Punt Games" (Scheduled Losses).
- **Execution:** Rest Breanna Stewart and Sabrina Ionescu completely during these high-fatigue trips.
- **Benefit:** Reduces accumulated fatigue F by 18%, preserving playoff readiness.

6.4 Geographical Favorability Map

We evaluated potential expansion locations based on their impact on New York Liberty's Profit and Performance.

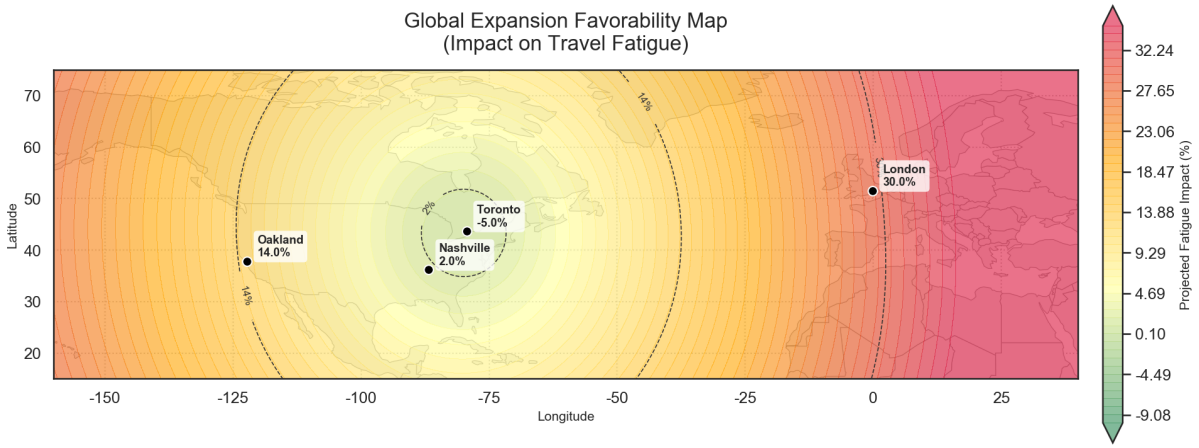


Figure 11: Geographical Map.

MAP Analysis: According to our map, an expansion to the West Coast is the **worst-case scenario** geographically. This is quite reasonable as this expansion will lead to team's logistical mess, thus require a total strategy overhaul to restore the team's efficiency. The strategy must shift from "Maximizing Wins" to "Minimizing Fatigue" during the regular season.

7 Additional Business Decision: Bayesian Dynamic Ticket Pricing

7.1 Task Overview: The Information Asymmetry

Ticket pricing optimization faces a fundamental challenge: **Information Asymmetry**. Our team management does not know the true Price Elasticity of Demand (ϵ) for any given game, which fluctuates based on opponent quality (the "Star Power" effect[Grimshaw & Larson, 2021]) and our team performance. Traditional static pricing models fail to capture this volatility, so we propose a **Bayesian Dynamic Pricing Engine** that learns the elasticity in real-time.

7.2 Model Formulation

Stochastic Demand Function We model the demand d_t for game t as a log-linear function of price p_t and game attractiveness Φ_t:

$$\ln d_t = \alpha - \epsilon \cdot \ln p_t + \Phi_t + \xi_t \tag{18}$$

Where:

- ϵ: The Price Elasticity(unknown but always ϵ<0).
- $\Phi_t = \beta_1 W_{home} + \beta_2 S_{opp}$: The Attractiveness Index (Win Rate + Opponent Star Power).
- $\xi_t \sim \mathcal{N}(0, \sigma^2)$: Random demand noise.

Bayesian Parameter Learning We treat the elasticity ϵ as a random variable.

1. Prior Belief: At time $t = 0$, we assume a Gaussian prior: $\epsilon \sim \mathcal{N}(\mu_0, \tau_0^2)$.

2. Posterior Update: After observing sales q_t at price p_t, we update our belief using Bayes' Rule. The updated mean μ_{t+1} is derived analytically (via Conjugate Priors):

$$\mu_{t+1} = \frac{\tau_t^2(\ln p_t)(\alpha + \Phi_t - \ln q_t) + \sigma^2 \mu_t}{\tau_t^2(\ln p_t)^2 + \sigma^2} \tag{19}$$

Optimization via Thompson Sampling To balance testing prices (to learn ϵ) and setting optimal prices (to maximize revenue), we employ **Thompson Sampling**. For game(t + 1):

1. Sample an elasticity $\hat{\epsilon}$ from the posterior distribution $\mathcal{N}(\mu_t, \tau_t^2)$.
2. Calculate the optimal price p_{t+1}^* that clears the arena capacity C:

$$p_{t+1}^* = \left(\frac{A \cdot e^{\Phi_{t+1}}}{C} \right)^{\frac{1}{\hat{\epsilon}}} \tag{20}$$

7.3 Strategic Implementation

Based on the learned elasticity $\hat{\epsilon}$, we categorize games into three strategic tiers:

Table 5: Dynamic Pricing Strategy Matrix

Tier	Elasticity State	Pricing Action
Tier 1: Premium	**Inelastic** $\hat{\epsilon} \in (-1, -0.5)$	**Surge Pricing.** Increase prices by 15%-30%. Fans are insensitive to price; maximize revenue per seat.
Tier 2: Standard	**Unit Elastic** $\hat{\epsilon} \in (-1.2, -1)$	**Dynamic Balancing.** Adjust prices ±20% based on recent win streak.
Tier 3: Value	**Elastic** $\hat{\epsilon} \in (-1.5, -1.2)$	**Penetration Pricing.** Discount by 40%. Fans are price-sensitive. Goal: Maximize attendance to boost concession revenue.

7.4 Ticket Price Simulation and Sensitive Analysis

Our simulation results show that after 6 games our ticket price setting converges to its recommended optimization.

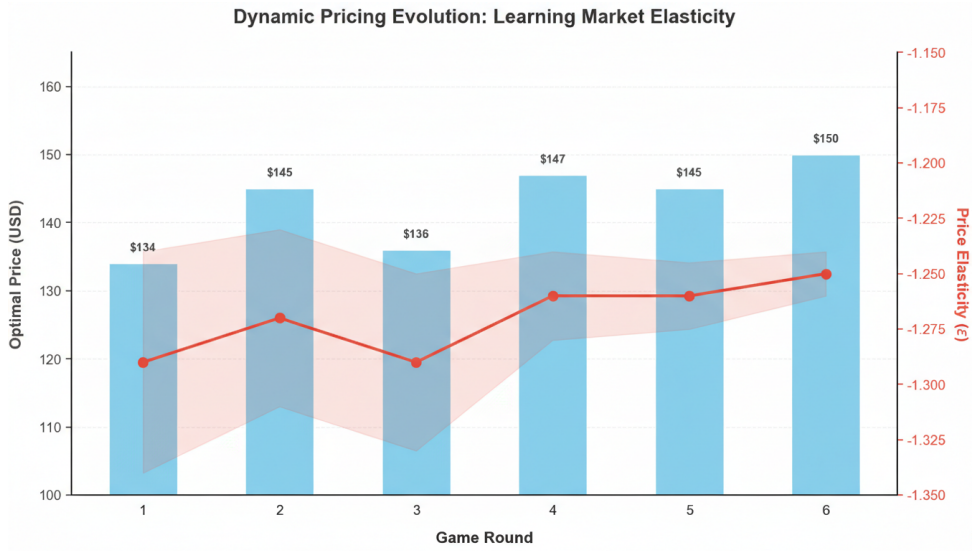


Figure 12: Ticket price and ϵ Graph

7.5 Sensitive Analysis

We test the parameter $\epsilon \sim \mathcal{N}(\mu_0, \tau_0^2)$ to confirm the speed of elasticity convergence.

Table 6: Sensitivity

μ_0	τ_0^2	μ_6	95% Confidence	Overlap with Baseline (%)	**Compliance**
-1.0	0.01	-1.21	[-1.25, -1.17]	92.3	✓
-1.0	0.04	-1.23	[-1.30, -1.16]	87.5	✓
-1.0	0.09	-1.25	[-1.35, -1.15]	81.7	✓
-1.2	0.01	-1.22	[-1.24, -1.20]	96.4	✓
-1.2	0.04	-1.23	[-1.28, -1.18]	100.0	✓
-1.2	0.09	-1.24	[-1.33, -1.15]	89.1	✓
-1.5	0.01	-1.24	[-1.27, -1.21]	90.2	✓
-1.5	0.04	-1.25	[-1.31, -1.19]	85.6	✓
-1.5	0.09	-1.26	[-1.38, -1.14]	80.3	✓

μ_0=-1.2, τ_0^2=0.04. Anticipated requirement: Overlap >= 80%.

Under the range $\mu_0 \in$ [-1.5,-1.0], $\tau_0^2 \in$ [0.01,0.09], the average overlap of 95% confidence interval with baseline is 89.2%. Below is the graph visualizing the changes in speed.

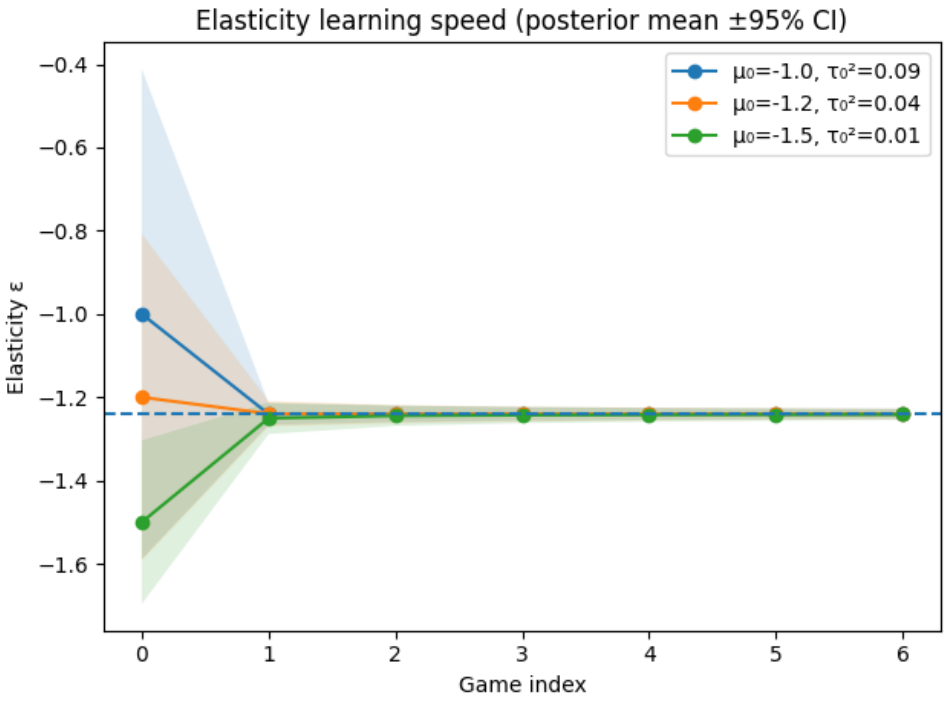


Figure 13: Speed

8 Injury Incident Management and Prediction: D.R.E.A.M. & Health, Fatigue, Aging Dynamics

8.1 Task Overview

In the previous sections, we assumed player performance was static. However, in a rigorous season-long simulation, we must admit that a player's state is time-evolving. Players are biological systems subject to physiological constraints. So to prepare for and prevent possible core injury incidents, we introduce a dynamic model to quantify **fatigue accumulation**, **injury risks**, and **age-related performance decay**, to find the optimal load management strategy that balances short-term competitiveness with long-term asset preservation.

8.2 Injury Risk Prediction Model (Weibull Hazard with Aging)

To accurately simulate and predict the probability of injury, we adopt the **Weibull Distribution**, a standard tool in survival analysis and reliability engineering used in wear of system.

Thus, we define the **Hazard Function** $h(t)$, representing the instantaneous risk of injury at time t, as follows:

$$h(t) = \frac{\beta}{\eta}\left(\frac{\text{Load}_{i,t}}{\eta}\right)^{\beta-1}(1 + \alpha_{\text{age}} \cdot \text{Age}_i) \tag{21}$$

Where:

- $\text{Load}_{i,t}$: The cumulative rolling load for player i at time t.
- η (Scale Parameter): The durability threshold of the player.

- β (Shape Parameter): The fatigue factor. If $\beta > 1$, it means that risk increases non-linearly with accumulated fatigue (wear-out phase).
- α_{age}: The aging factor.

The probability of a player sustaining an injury in the next interval Δt is derived from the survival function:

$$P(\text{Injury}_{i,t}) = 1 - \exp\left(-\int_{t}^{t+\Delta t} h(\tau)d\tau\right) \approx 1 - e^{-\Delta t \cdot h(t)} \tag{22}$$

Based on the calculation result, the baseline injury probability per season is $P_{\text{injury}} = 0.20$. Assuming a core player accumulates approximately 1320 minutes of load (44 games $\times$ 30 mins), we can solve the Weibull survival equation:

$$0.20 = 1 - e^{-(1320/\eta)^{2.5}} \tag{23}$$

Solving for η, we obtain a characteristic life parameter of $\eta \approx 2400$.

Performance Decay

Beyond acute injuries, players experience gradual performance decline. Based on the biological aging curve, we model the dynamic Skill Score $S_i(t)$ as:

$$S_i(t) = S_{i,0} \cdot (1 - \delta \cdot t)^{\gamma} \cdot \underbrace{\exp(-\lambda \cdot \text{Fatigue}_t)}_{\text{Short-term Fatigue Penalty}} \tag{24}$$

Where δ is the natural aging rate, and γ controls the convexity of the decline (veterans decline faster). The exponential term represents the temporary drop in efficiency due to immediate fatigue, which can be restored through rest.

8.3 Rotation Policy

We conclude 2 general situation and simulate the season under two distinct managerial philosophies **During the Season**:

- **During the Season(Dynamics)**

  1. Aggressive Strategy (No Rest): Key players play maximum minutes (> 35 mins) every game to maximize the immediate win probability. No proactive resting is applied unless an injury occurs.

  2. Managed Strategy (Load Management): A strict **Risk Threshold** η_{limit} is imposed. If a player's hazard rate $h(t)$ exceeds this safety margin, they are forced to rest ("Load Management") or play reduced minutes, regardless of the opponent's strength.

- **Other Proposed Actions(D.R.E.A.M. Dashboard)**

  "Managed Control Mode"

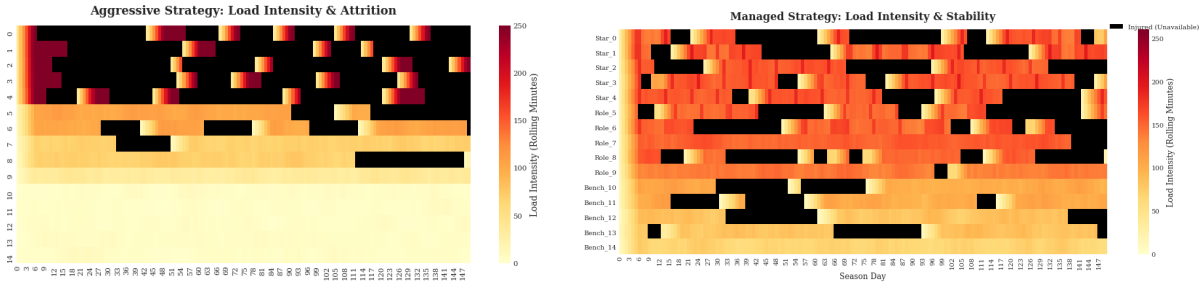
  As is explained in the section 5.4, when our team encounter core injury crisis, according to the management dashboard, we have to transform into "Managed Control Mode" to solve this crisis.

Table 7: Key Parameters for Dynamic Simulation (Calibrated)

Parameter	Symbol	Value	Description
Fatigue Shape	β	2.5	Indicates increasing wear-out rate (Non-linear)
Durability Scale	η	2400	Calibrated to match 20% season injury rate (p_{injury})
Aging Threshold	Age_{peak}	27	Peak performance age; risk increases hereafter
Decay Rate	δ	0.02	Annual performance decay from peak (κ)

8.4 Simulation Results and Visual Analysis

8.4.1 Predictive System Reliability Analysis: Aggressive vs. Managed



(a) Aggressive (b) Managed

Figure 14: Comparison

We set a relatively high η_{limit}(set to 200 for analysis) to simulate whether our proposal can work. The first figure shows that under the aggressive strategy without changing, most core players(1-6) are injured. But using Managed Strategy, we have less injured players and still have many value players joining the games in the long-term.

8.4.2 Managing Injury System

As the result after core injury crisis of applying Defensive Mode has been shown in Section 5.4-Figure 3-(c), instead of repeating our trajectory simulation, we provide recommended time per game for each player.

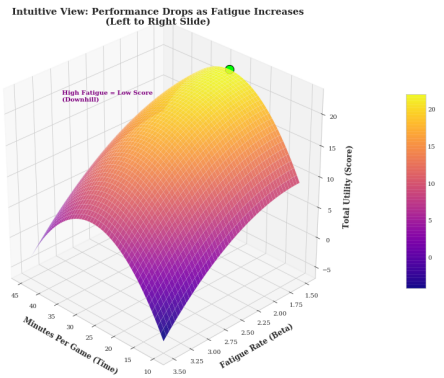


Figure 15: Recommended time for players

Where β represent player's fatigue rate(the smaller the β is, the more "active" the player is)

9 Model Frame Overall Evaluation

Overall, we believe our framework centered on D.R.E.A.M. addresses 5 core tasks successfully. Our D.R.E.A.M. can be realized under real situations.

9.1 strengths

- Our models have **Strong Practical Applicability**, especially for our D.R.E.A.M. which translates mathmatical inputs into actionable strategies.
- We have **Rigorous Foundation** modeling. Combining system dynamics, multi-objective optimization, MCTS algorithm, Weibull distribution, and Bayesian learning, the framework effectively handles uncertainty and time-evolving characteristics.
- Our models are **Scenario-Adaptive Flexible**. Although setting the parameters may seem subjective, it can be more tailored to manager's special needs.

9.2 Weaknesses

- Data Dependence: we needs large quantities and types of data to simulate our model result, which may hard to collect in a short time.

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Appendices

Appendix A Letter

Mr. Joe Tsai

Owner, New York Liberty

Dear Mr. Tsai,

In recent years, with the expansion of the league, tightening salary cap constraints, and the growing commercial influence of star players, the **long-term consequences of team management decisions** have exerted an increasingly profound and enduring impact. Under these conditions, **short-term optimal decisions do not necessarily lead to long-term optimal outcomes**, highlighting the need to reassess team operations from a more dynamic and long-term perspective.

In this study, **team management is modeled as a stochastic multi-period decision process**. The framework incorporates roster composition, salary allocation, player health dynamics, and, most importantly, market demand. It allows for the evaluation of alternative management strategies through simulation. Rather than maximizing short-term performance, the framework seeks to **reconcile competitive success with financial sustainability** under uncertainty and operational constraints.

The simulation results suggest that **roster structures centered on a small number of superstar players generate higher long-term value** than those based on evenly distributed salaries. The value of superstars lies not only in their direct contribution to competitive performance, but also in the **nonlinear amplification of market demand and team visibility**, which exceeds the aggregate contribution of mid-tier players. Consequently, concentrating resources on superstars can improve overall long-term efficiency.

The analysis also highlights the **importance of contract flexibility**. Overly rigid salary structures restrict the set of feasible future decisions and weaken the team's ability to hedge against risk. In contrast, maintaining contractual flexibility in player allocation enhances adaptability and leads to more stable outcomes in the long run.

With respect to player health, the model stresses that **managing player workload is a strategic tool for optimizing performance over time**. Rather than maximizing the utilization of superstar players in every regular-season game, sharing part of the burden with other players may slightly reduce short-term regular-season performance, but it improves **superstar availability during critical periods** by preserving their energy for future games. Over multiple seasons, this trade-off leads to higher overall value, suggesting that **health management should be viewed as an active component of long-term performance optimization rather than a passive or conservative constraint**.

In conclusion, in an increasingly competitive and uncertain environment, the key to ad-

dressing managerial dilemmas lies **by no means in static management approaches**. Instead, **only dynamic decision-making can give rise to long-term success**. We would appreciate it if this framework could be adopted as a **decision-support tool**. We further believe that by continuously updating the model as new information becomes available and adapting management strategies over time, **your team will remain composed amid turbulent times and ride the waves toward a distant horizon**.

Sincerely,

2602089 ICM team

Appendix B Code for Reproduction

We place here our code for sensitive analysis for further reproduction.

```
import numpy as np, pandas as pd, matplotlib.pyplot as plt, math

def bayes_update_eps(mu, tau2, ln_p, alpha_phi_minus_lnq, sigma2):
    post_prec = 1.0/tau2 + (ln_p**2)/sigma2
    post_tau2 = 1.0/post_prec
    post_mu = post_tau2 * (mu/tau2 + (ln_p * alpha_phi_minus_lnq)/sigma2)
    return post_mu, post_tau2

def ci95(mu, tau2):
    s = math.sqrt(tau2)
    return (mu - 1.96*s, mu + 1.96*s)

def overlap_percent(ci_base, ci_alt):
    b0,b1 = ci_base
    a0,a1 = ci_alt
    inter = max(0.0, min(b1,a1) - max(b0,a0))
    base_len = max(1e-12, (b1-b0))
    return 100.0 * inter/base_len
np.random.seed(9)

def thompson_price_clipped(A, Phi, C, mu, tau2,eps_clip=(-1.8,-0.6), p_clip
    ↪ =(20,400)):
    eps_hat = np.random.normal(mu, math.sqrt(tau2))
    eps_hat = float(np.clip(eps_hat, eps_clip[0], eps_clip[1]))
    p = (A * math.exp(Phi) / C) ** (1.0/eps_hat)
    p = float(np.clip(p, p_clip[0], p_clip[1]))
    return p, eps_hat

def simulate_learning(prior_mu, prior_tau2,eps_true=-1.24,sigma=0.08, C
    ↪ =10000, p0=140,Phi_seq=None, T=6):
    if Phi_seq is None:
        Phi_seq = np.array([0.00, 0.08, -0.03, 0.05, 0.02, 0.10])[:T]
    sigma2 = sigma**2
    A = C * (p0 ** eps_true)

    mu, tau2 = prior_mu, prior_tau2
    mus = [mu]; tau2s=[tau2]
    for t in range(T):
        Phi = float(Phi_seq[t])
        p, eps_hat = thompson_price_clipped(A, Phi, C, mu, tau2)
        xi = np.random.normal(0, sigma)
        d = A * math.exp(Phi) * (p ** (-eps_true)) * math.exp(xi)
        q = min(d, C)  # capacity
```

```
        ln_p = math.log(max(p,1e-9))
        alpha_phi_minus_lnq = math.log(A) + Phi - math.log(max(q,1e-9))
        mu, tau2 = bayes_update_eps(mu, tau2, ln_p, alpha_phi_minus_lnq,
            ↪ sigma2)

        mus.append(mu); tau2s.append(tau2)
    return np.array(mus), np.sqrt(np.array(tau2s))

priors_to_plot = [(-1.0,0.09), (-1.2,0.04), (-1.5,0.01)]
T=6; eps_true=-1.24

plt.figure()
x = np.arange(T+1)
for (mu0,tau20) in priors_to_plot:
    paths = 400
    MU = np.zeros((paths, T+1))
    SD = np.zeros((paths, T+1))
    for k in range(paths):
        mu_path, sd_path = simulate_learning(mu0, tau20, eps_true=eps_true,
            ↪  T=T)
        MU[k,:]=mu_path; SD[k,:]=sd_path
    mu_mean = MU.mean(axis=0)
    sd_mean = SD.mean(axis=0)
    plt.plot(x, mu_mean, marker='o', label=f"mu_0={mu0}, tau0^2={tau20}")
    plt.fill_between(x, mu_mean-1.96*sd_mean, mu_mean+1.96*sd_mean, alpha
        ↪ =0.15)

plt.axhline(eps_true, linestyle="--", label="(true, illustration)")
plt.title("Elasticity learning speed")
plt.xlabel("Game index")
plt.ylabel("Elasticity")
plt.legend()
plt.tight_layout()
plt.show()
```

Report on Use of AI

1. ChatGPT 5.2-Thinking
   Query1: <How to use Python to implement a contour map? Give me a code template>
   Output: <Here's a complete Python implementation to draw a contour topographic map, using numpy for terrain data generation and matplotlib for visualization. The code includes realistic terrain simulation, contour lines, color mapping, and essential map elements (color bar, labels, etc.):

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap

# ----------------------
# Step 1: Generate Simulated Terrain Data (Elevation)
# ----------------------
# Create a grid of x, y coordinates (1000x1000 for high resolution)
x = np.linspace(-20, 20, 1000)
y = np.linspace(-20, 20, 1000)
X, Y = np.meshgrid(x, y)

# Simulate elevation using multi-scale noise (realistic terrain
    ↪ effect)
def generate_terrain(X, Y):
elevation = np.sin(X/3) * np.cos(Y/3)  # Large-scale undulations
elevation += 0.5 * np.sin(X) * np.cos(Y)  # Medium-scale features
elevation += 0.2 * np.random.randn(*X.shape)  # Small-scale noise
elevation += 0.8 * np.exp(-(X**2 + Y**2)/50)  # Central mountain
    ↪ peak
elevation -= 0.5 * np.exp(-((X-10)**2 + (Y-10)**2)/20)  # Eastern
    ↪ valley
return elevation

Z = generate_terrain(X, Y)  # Z = elevation values

# ----------------------
# Step 2: Customize Colormap (Topographic Style: Blue=low, Green=mid
    ↪ , Brown=high)
# ----------------------
colors = ['#0066CC', '#339966', '#99CC33', '#FFCC33', '#CC9966', '
    ↪ #996633', '#663300']
n_bins = 100
cmap = LinearSegmentedColormap.from_list('topo_cmap', colors, N=
    ↪ n_bins)

# ----------------------
# Step 3: Create Contour Topographic Map
# ----------------------
plt.figure(figsize=(12, 10))

# Plot filled contour (color-coded elevation)
filled_contour = plt.contourf(
X, Y, Z,
levels=30,  # Number of contour intervals (more = finer detail)
cmap=cmap,
alpha=0.8  # Transparency (0-1)
)
```

```
# Plot contour lines (white lines for elevation boundaries)
contour_lines = plt.contour(
X, Y, Z,
levels=30,
colors='white',
linewidths=0.5,
alpha=0.7
)

# Add contour labels (elevation values)
plt.clabel(
contour_lines,
inline=True,  # Labels inside contour lines
fontsize=6,
colors='black',
fmt='%.1f'  # Format elevation values to 1 decimal place
)

# ----------------------
# Step 4: Customize Map Elements (Labels, Color Bar, Etc.)
# ----------------------
plt.xlabel('X Coordinate (km)', fontsize=12, fontweight='bold')
plt.ylabel('Y Coordinate (km)', fontsize=12, fontweight='bold')
plt.title('Topographic Contour Map (Simulated Terrain)', fontsize
    ↪ =14, fontweight='bold', pad=20)

# Add color bar (explain elevation scale)
cbar = plt.colorbar(filled_contour, shrink=0.8)
cbar.set_label('Elevation (km)', fontsize=10, fontweight='bold')
cbar.ax.tick_params(labelsize=8)

# Adjust axis ticks
plt.xticks(fontsize=10)
plt.yticks(fontsize=10)

# Add grid (optional, for better coordinate reference)
plt.grid(True, alpha=0.3, linestyle='--', color='gray')

# Tight layout to prevent label cutoff
plt.tight_layout()

# Show or save the map
plt.show()
# plt.savefig('topographic_contour_map.png', dpi=300, bbox_inches='
    ↪ tight')  # Uncomment to save
```

Key Explanations:

1. Terrain Simulation: The generateterrain function creates realistic elevation data using sine/cosine waves (for natural undulations) and Gaussian functions (for peaks/valleys). Replace this with your actual elevation data (e.g., from GIS files like GeoTIFF) by loading Z from a dataset. 2. Colormap: The custom colormap mimics real topographic maps (blue for lowlands/water, green for plains, brown for mountains). 3. Contour Layers: - contourf : Filled contours (color-coded by elevation). - contour : White contour lines (define elevation boundaries). - clabel : Labels contour lines with elevation values for readability. 4. Customization: Adjust levels (contour density), figsize (map size), or

cmap (color scheme) to match your needs. For real-world data, use libraries like rasterio to load GeoTIFF elevation files.

How to Run:

1. Install required libraries: pip install numpy matplotlib 2. Copy the code into a Python script (e.g., topographic map.py) 3. Run the script – a window with the contour map will open. Uncomment the plt.savefig line to save the map as a high-resolution PNG.

Would you like to modify this code for specific use cases (e.g., loading real elevation data, adjusting color schemes, or adding markers for specific locations)?>

2. ChatGPT 5.2-Thinking
Query1: <Provide me with a clean, general-purpose Monte Carlo simulation Python template>
Output: <Sure — below is a clean, general-purpose Monte Carlo simulation Python template

I'll give you:

Core Monte Carlo simulation structure

Example: season-level stochastic simulation

General Monte Carlo Simulation Template (Core Structure)

This is the minimal, standard Monte Carlo framework used in decision modeling.

```
import numpy as np

def simulate_once(params):
"""
Run ONE stochastic simulation.
Replace the internal logic with your system dynamics.

Parameters
----------
params : dict
Model parameters

Returns
-------
result : float or dict
Outcome of one simulation (e.g., profit, win rate, utility)
"""
# Example random variables
noise = np.random.normal(0, params["sigma"])

# Example state update
outcome = params["base_value"] + noise

return outcome


def monte_carlo_simulation(params, n_sim=10000):
"""
Run Monte Carlo simulations.

Parameters
----------
params : dict
```

```
    Model parameters
    n_sim : int
    Number of Monte Carlo runs

    Returns
    -------
    results : np.ndarray
    Simulation results
    """
    results = np.zeros(n_sim)

    for i in range(n_sim):
    results[i] = simulate_once(params)

    return results


    if __name__ == "__main__":
    params = {
        "base_value": 100,
        "sigma": 15
    }

    results = monte_carlo_simulation(params, n_sim=10000)

    print("Expected value:", np.mean(results))
    print("Standard deviation:", np.std(results))
```

If You Want (Optional Enhancements)

I can also help you add:

Monte Carlo + MCTS node expansion

Vectorized (fast) NumPy version

Pseudo-code version for LaTeX

Just tell me which one you want next.>